

Methodology 2 — Frank-Wolfe on the polytope

Linear minimisation oracle, no projections

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Draft outline — content to come.

The Frank-Wolfe iteration

At iterate $t^{(r)}$:

1. **Linear minimisation oracle (LMO):**

$$s^{(r)} \in \arg \min_{u \in \mathcal{T}_k} \langle g(t^{(r)}), u \rangle \iff s^{(r)} = \text{vertex with the } k \text{ smallest entries of } g(t^{(r)}).$$

2. **Step:** $t^{(r+1)} = (1 - \alpha) t^{(r)} + \alpha s^{(r)}$.

3. **Iterate** until $t^{(r)}$ converges to a vertex.

Why no projections

The cardinality polytope is a *combinatorial* set; projecting onto it is itself a hard problem. Frank-Wolfe avoids the projection — every step just requires the LMO, which is a sort.

Vertex-approach guarantee

$$\|t^{(r)} - s^*\|_\infty \leq (1 - \alpha)^r \|t^{(0)} - s^*\|_\infty.$$

Linear convergence to the chosen vertex once the LMO direction settles.

Placeholder for visual: a small 3D figure of FW iterates marching toward a vertex of the polytope.

$\leftarrow$ Boolean relaxation $\cdot$ Next $\rightarrow$ Homotopy